

Unit 12

NITROGEN AND SULFUR

46 items · 109 marks

Syllabus sections in this unit

12.1 Nitrogen and sulfur

Unit 12: Nitrogen and sulfur

1. M/J 26/P21/Q2(a)

Context

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

(a) Write an overall equation for stage 3.

[1]

2. M/J 26/P21/Q2(c)(i)

Context

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

Context: part (c) (introduction)

(c) CaCO_3 and O_2 are added to the waste gases from the Contact process before they escape. This prevents the release of SO_2 into the atmosphere.



(i) State **one** environmental consequence of the release of $\text{SO}_2(\text{g})$ into the atmosphere.

[1]

3. M/J 26/P24/Q2(a)(ii)

(ii) Explain the lack of reactivity of nitrogen.

[2]

4. MAR 26/P22/Q2(c)

Context

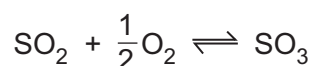
2 Nitrogen monoxide, NO, is an atmospheric pollutant.**(c) (i)** State a natural process that creates atmospheric NO.

[1]

(ii) Photochemical smog forms when NO reacts with unburned hydrocarbons.

Identify the major component of photochemical smog.

[1]

(iii) NO has a catalytic role in the oxidation of atmospheric SO₂.

Show, using relevant equations, the catalytic role that NO has in this oxidation.

[2]

5. M/J 25/P23/Q3(a)(ii)

Context: part (a) (introduction)

3 (a) Different hydrocarbon mixtures produced from fractional distillation of crude oil have different uses.**(ii)** Describe how photochemical smog is produced during the combustion of petrol in an internal combustion engine.

[2]

6. O/N 24/P22/Q3(a)(i)-(iii)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (a) (introduction)

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.**(i)** Identify **one** natural and **one** man-made occurrence of nitrogen oxides in the atmosphere.

natural

man-made

[2]

(ii) Write an equation to describe the role of NO₂ in the direct formation of acid rain.

[1]

(iii) Peroxyacetyl nitrate, PAN, is a component of photochemical smog.Describe how PAN forms from NO₂.

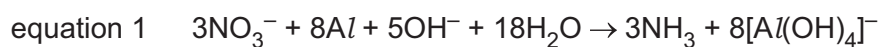
[1]

7. O/N 24/P22/Q3(b)(iii)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (b) (introduction)

(b) A common test for nitrates is the reaction with NaOH and Al. Equation 1 shows the reaction.**(iii)** NH_3 is a basic gas.Describe how NH_3 is able to act as a base.

[1]

8. O/N 24/P22/Q3(f)(ii)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (f) (introduction)

(f) At very high temperatures, phosphorus can form P_2 molecules. P_2 contains a triple bond, $\text{P}\equiv\text{P}$.**(ii)** The bond energy of $\text{P}\equiv\text{P}$ is 485kJ mol^{-1} . The bond energy of $\text{N}\equiv\text{N}$ is 944kJ mol^{-1} . Compare the reactivity of P_2 and N_2 . Explain your answer.

[1]

9. M/J 24/P23/Q1(a)(i)**1 (a) (i)** Explain the lack of reactivity of nitrogen gas, $\text{N}_2(\text{g})$.

[2]

10. M/J 24/P23/Q2(d)

Context

2 $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.

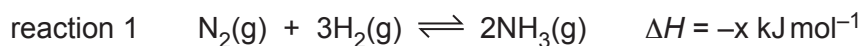


Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

(d) When oxides of nitrogen escape into the atmosphere they may be involved in:

- formation of acid rain from sulfur dioxide
- formation of photochemical smog.

(i) Identify the role of NO and NO_2 in the formation of H_2SO_4 from SO_2 in the atmosphere to produce acid rain.

Use relevant equations to support your answer.

[3]

(ii) Outline how NO and NO_2 may contribute to the formation of photochemical smog.

[2]

11. M/J 24/P23/Q4(b)(ii)

Context: part (b) (introduction)

(b) An exothermic reaction occurs when $\text{NH}_4^+(\text{aq})$ is added to $\text{OH}^-(\text{aq})$.

(ii) Construct an ionic equation for the reaction of NH_4^+ and OH^- .

[1]

12. MAR 24/P22/Q3(a)

Context

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

(a) Give **two** reasons to explain the lack of reactivity of nitrogen.

1

2

[2]

13. MAR 24/P22/Q3(b)

Context

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

(b) N_2 can react with oxygen in an internal combustion engine to form a mixture of NO and NO_2 . Fig. 3.1 shows a reaction scheme involving N_2 .

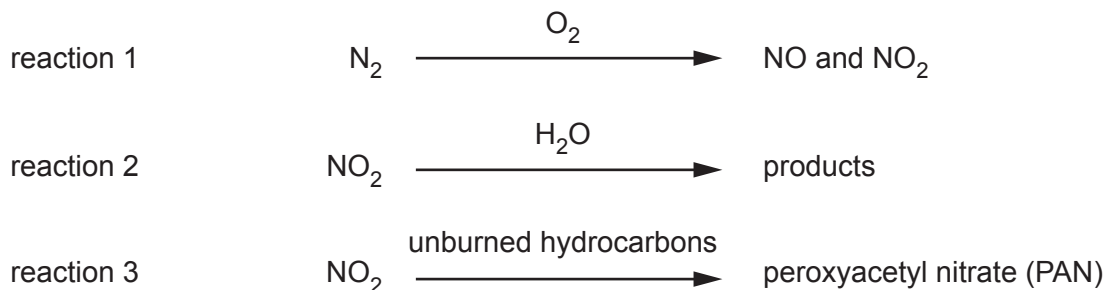


Fig. 3.1

(i) Write an equation to show the formation of a mixture of NO and NO_2 in reaction 1.

[1]

(ii) Give the formulae of the products of reaction 2.

[1]

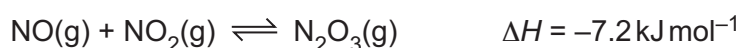
(iii) State **one** environmental consequence of reaction 3.

[1]

14. O/N 23/P22/Q2(d)

Context

2 NO and NO_2 react at 25 °C to give N_2O_3 as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

(d) Identify one natural process and one man-made process that cause the formation of atmospheric NO and NO_2 .

natural process

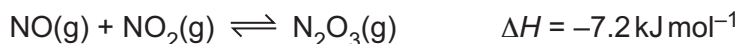
man-made process

[2]

15. O/N 23/P22/Q2(e)(ii)-(iii)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

Context: part (e) (introduction)

- (e)** NO₂ is a brown gas that can be used to form nitric acid.

- (ii)** NO₂ has a catalytic role in the oxidation of atmospheric sulfur dioxide.

Write equations to show the catalytic role of NO₂ in this oxidation.

[2]

- (iii)** State **one** environmental consequence of the oxidation of atmospheric sulfur dioxide.

[1]

16. M/J 23/P21/Q5(b)

Context: part (a) (introduction)

- 5** **(a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.

- (b)** When petrol is burned in an internal combustion engine, oxides of nitrogen are released into the atmosphere. Oxides of nitrogen are responsible for the formation of acid rain.

- (i)** Suggest the conditions required for the production of oxides of nitrogen during combustion of **M** in an internal combustion engine. Use an appropriate equation in your answer.

[2]

- (ii)** Describe how acid rain is formed in the atmosphere in the presence of oxides of nitrogen and SO₂. Identify the role of the oxides of nitrogen in this process. Include **all** relevant equations.

[3]

- (iii)** State **one** other type of air pollution that is caused by the production of oxides of nitrogen in an internal combustion engine.

[1]

17. O/N 22/P22/Q1(c)(ii)

Context

1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

Context: part (c) (introduction)

(c) NH_4^+ is a Brønsted–Lowry acid.

(ii) When $\text{NH}_4^+(\text{aq})$ is heated with $\text{NaOH}(\text{aq})$, a pungent gas is produced.

Write an ionic equation for this reaction.

[1]

18. M/J 22/P21/Q2(b)

Context

2 Nitrogen molecules, $\text{N}_2(\text{g})$, contain two atoms attracted to each other by a triple covalent bond.

(b) Nitrogen oxides, NO_2 and NO , are produced in internal combustion engines. Release of these gases into the atmosphere leads to the formation of photochemical smog.

(i) Outline how nitrogen oxides are involved in the formation of photochemical smog.

[2]

(ii) Construct an equation to demonstrate how a catalytic converter reduces the amount of nitrogen oxide gases released into the atmosphere.

[1]

19. M/J 22/P21/Q2(c)

Context

2 Nitrogen molecules, $\text{N}_2(\text{g})$, contain two atoms attracted to each other by a triple covalent bond.

(c) $\text{N}_2(\text{g})$ is very unreactive. It is difficult to make ammonia, $\text{NH}_3(\text{g})$, directly from its elements but it can be made from $\text{NH}_4\text{Cl}(\text{s})$.

Identify a reagent and the conditions required to make $\text{NH}_3(\text{g})$ from $\text{NH}_4\text{Cl}(\text{s})$.

[1]

20. MAR 22/P22/Q2(d)(i)-(ii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (d) (introduction)

(d) SO_2 and SO_3 are found in the atmosphere.

The oxidation of SO_2 to SO_3 in the atmosphere is catalysed by NO_2 .

The first step of the catalytic oxidation is shown in equation 1.



(i) Construct an equation to show how NO_2 is regenerated in the catalytic oxidation of SO_2 .

[1]

(ii) NO_2 can also react with unburned hydrocarbons to form photochemical smog.

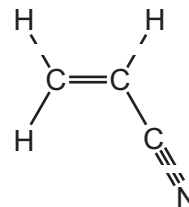
State the product of this reaction that contributes to photochemical smog.

[1]

21. MAR 21/P22/Q3(a)(ii)

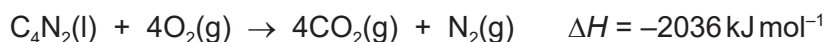
Context

3 Compounds **P**, **Q** and **R** have all been found in the atmosphere of one of Saturn's moons.

P**Q****R**

Context: part (a) (introduction)

(a) The equation for the complete combustion of **P**, $\text{C}_4\text{N}_2(\text{l})$, is shown.



(ii) One of the products of the complete combustion of **P** is nitrogen gas, $\text{N}_2(\text{g})$.

Explain the lack of reactivity of nitrogen.

[1]

22. O/N 20/P21/Q2(b)(iii)

Context

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

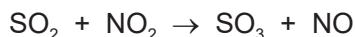
Context: part (b) (introduction)

(b) The two most common oxides of sulfur are SO_2 and SO_3 .

When SO_2 dissolves in water, a small proportion of it reacts with water to form a weak Brønsted-Lowry acid.

(iii) SO_2 reacts with NO_2 in the atmosphere to form SO_3 and NO .

NO is then oxidised in air to form NO_2 .



State the role of NO_2 in this two-stage process.

[1]

23. O/N 20/P23/Q2(b)(iii)

Context

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

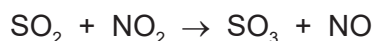
Context: part (b) (introduction)

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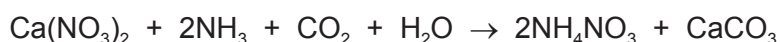
State the role of NO_2 in this two-stage process.

[1]

24. M/J 20/P21/Q4(a)

Context

4 Calcium nitrate, $\text{Ca}(\text{NO}_3)_2$, reacts with ammonia, carbon dioxide and water to form a mixture of ammonium nitrate and calcium carbonate.



(a) Explain why ammonia is described as a Brønsted-Lowry base in this reaction.

[1]

The product mixture can then be added to soil.

25. O/N 19/P22/Q1(c)(ii)

Context

- 1** In the Periodic Table, the p block contains elements whose outer electrons are found in the p subshell.

Context: part (c) (introduction)

- (c)** Sulfur-containing compounds, such as $\text{C}_2\text{H}_5\text{SH}$, are found in fossil fuels, and produce SO_2 when they are burned.

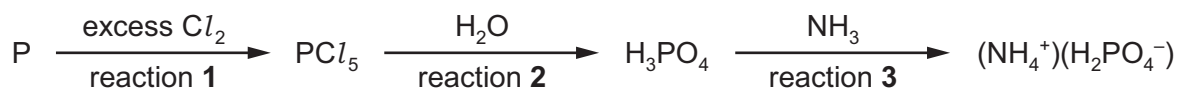
- (ii)** State why the presence of SO_2 in the atmosphere has environmental consequences. Describe **one** of the consequences on the environment.

[2]

26. O/N 19/P22/Q3(b)(i)

Context

- 3** A series of reactions for phosphorus and its compounds is shown.



Context: part (b) (introduction)

- (b)** Reaction **3** is a neutralisation reaction in which NH_3 acts as a base.

- (i)** Explain how NH_3 acts as a base in reaction **3**.

[1]

27. M/J 19/P23/Q4(a)

Context

- 4** Release of sulfur dioxide, SO_2 , into the atmosphere causes acid rain.

- (a)** Explain why high levels of SO_2 may be found in the atmosphere near power stations that burn fossil fuels.

[2]

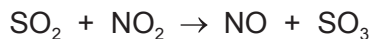
28. M/J 19/P23/Q4(b)(ii)-(iv)

Context

4 Release of sulfur dioxide, SO_2 , into the atmosphere causes acid rain.

Context: part (b) (introduction)

(b) The SO_2 released can be converted in the atmosphere into sulfuric acid, H_2SO_4 , by reaction with nitrogen dioxide gas, NO_2 , and water, H_2O .



During one year, 1590 tonnes of SO_2 was released into the atmosphere by a fossil-fuel burning power station.

(ii) Describe how NO_2 is also produced by these power stations.

[2]

(iii) State one natural cause of NO_2 being formed in the atmosphere.

[1]

(iv) Explain why NO_2 can be described as a catalyst in the oxidation of atmospheric SO_2 .

[2]

29. MAR 19/P22/Q1(a)

Context

1 Nitrogen, N_2 , is the most abundant gas in the Earth's atmosphere and is very unreactive.

(a) State why N_2 is very unreactive.

[1]

30. MAR 19/P22/Q1(c)

Context

1 Nitrogen, N_2 , is the most abundant gas in the Earth's atmosphere and is very unreactive.

(c) (i) State the industrial importance of ammonia.

[1]

(ii) One method of producing NH_3 is by heating ammonium chloride, NH_4Cl , with CaO .



Explain why the reaction of NH_4Cl with CaO produces ammonia.

[2]

31. O/N 18/P21/Q1(c)(i)

Context

- 1** Iron pyrite, FeS_2 , has a yellow colour that makes it look like gold metal. The compound contains the ions Fe^{2+} and S_2^{2-} .

Context: part (c) (introduction)

- (c)** Iron pyrite is often called *fool's gold* because of its appearance. Impure samples of iron pyrite often contain a small amount of gold.

The gold can be obtained from impure iron pyrite. The impure iron pyrite is roasted in oxygen, to produce iron(III) oxide and sulfur dioxide. Gold does not react with oxygen.

- (i)** The sulfur dioxide produced during roasting would cause environmental consequences if released into the atmosphere.

State and explain **one** of these environmental consequences.

[2]

32. O/N 18/P21/Q2(a)(i)

Context: part (a) (introduction)

- 2 (a)** Nitrogen, N_2 , is an inert gas that makes up 78% of the Earth's atmosphere.

- (i)** Explain why nitrogen is inert.

[2]

33. O/N 18/P21/Q2(b)

- (b)** Nitrogen, N_2 , and oxygen, O_2 , react together in the air during lightning strikes to form nitrogen monoxide, NO .

- (i)** Explain why the reaction of N_2 and O_2 occurs during lightning strikes.

[1]

- (ii)** Write two equations to suggest how the NO formed reacts further to create nitric acid, HNO_3 .

1

2

[2]

34. O/N 18/P21/Q2(d)(ii)

Context: part (d) (introduction)

- (d) Some soils have compounds such as ammonium nitrate, calcium carbonate and calcium hydroxide added to them.
- (ii) When calcium hydroxide reacts with compounds containing the ammonium ion, NH_4^+ , a gas is produced.

State the identity of this gas and explain why the reaction occurs.

gas

explanation

[2]

35. O/N 18/P22/Q2(e)

Context

- 2 The table gives some data for elements in the third period and some of their compounds.

element	Na	Mg	Al	Si	P	S
type of bonding	metallic				covalent	covalent
formula of oxide					P_4O_{10}	SO_2
formula of chloride	NaCl	MgCl_2				SCl_2

- (e) SO_2 can be released into the atmosphere when fossil fuels containing sulfur are burnt.

State and explain one environmental consequence of the release of SO_2 into the atmosphere.

[2]

36. O/N 18/P23/Q1(c)(i)

Context

- 1 Iron pyrite, FeS_2 , has a yellow colour that makes it look like gold metal. The compound contains the ions Fe^{2+} and S_2^{2-} .

Context: part (c) (introduction)

- (c) Iron pyrite is often called *fool's gold* because of its appearance. Impure samples of iron pyrite often contain a small amount of gold.

The gold can be obtained from impure iron pyrite. The impure iron pyrite is roasted in oxygen, to produce iron(III) oxide and sulfur dioxide. Gold does not react with oxygen.

- (i) The sulfur dioxide produced during roasting would cause environmental consequences if released into the atmosphere.

State and explain **one** of these environmental consequences.

[2]

37. O/N 18/P23/Q2(a)(i)

Context: part (a) (introduction)

2 (a) Nitrogen, N_2 , is an inert gas that makes up 78% of the Earth's atmosphere.

(i) Explain why nitrogen is inert.

[2]

38. O/N 18/P23/Q2(b)

(b) Nitrogen, N_2 , and oxygen, O_2 , react together in the air during lightning strikes to form nitrogen monoxide, NO .

(i) Explain why the reaction of N_2 and O_2 occurs during lightning strikes.

[1]

(ii) Write two equations to suggest how the NO formed reacts further to create nitric acid, HNO_3 .

1

2

[2]

39. O/N 18/P23/Q2(d)(ii)

Context: part (d) (introduction)

(d) Some soils have compounds such as ammonium nitrate, calcium carbonate and calcium hydroxide added to them.

(ii) When calcium hydroxide reacts with compounds containing the ammonium ion, NH_4^+ , a gas is produced.

State the identity of this gas and explain why the reaction occurs.

gas

explanation

[2]

40. M/J 18/P22/Q3(d)(ii)-(iii)

Context

3 Most vehicle fuels contain hydrocarbons obtained from crude oil.

Context: part (d) (introduction)

(d) Vehicle fuels are treated to remove sulfur. If sulfur is present in a fuel when it is burned, SO_2 is produced and may be released into the atmosphere where it can form acid rain.

(ii) NO_2 is involved in the production of acid rain from SO_2 .

Give **two** equations which describe how acid rain is formed by the action of NO_2 with SO_2 .

[2]

(iii) NO_2 is described as a catalyst during this process.

Explain, with the use of an appropriate equation, why NO_2 is described as a catalyst.

[2]

41. M/J 17/P22/Q2(a)

Context

2 Nitrogen gas, N_2 , is very unreactive.

(a) Explain why nitrogen gas is so unreactive.

[2]

42. M/J 17/P22/Q2(b)

Context

2 Nitrogen gas, N_2 , is very unreactive.

(b) Despite the low reactivity of N_2 , oxides of nitrogen occur in the atmosphere through both natural and man-made processes.

(i) Explain why oxides of nitrogen can be produced by internal combustion engines.

[2]

(ii) State and explain, using a suitable equation, how oxides of nitrogen produced by internal combustion engines can be prevented from reaching the atmosphere.

[2]

(iii) State the role of nitrogen dioxide, NO_2 , in the formation of acid rain by oxides of sulfur. Write suitable equations to explain this role.

role

equation 1

equation 2

[3]

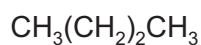
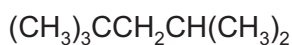
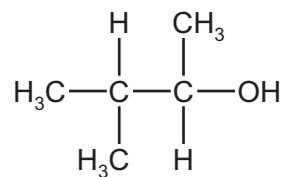
(iv) Suggest an equation to show how NO_2 can contribute **directly** to acid rain.

[1]

43. MAR 16/P22/Q4(c)

Context

- 4 The following compounds were all found to be components of a sample of petrol.

**G****H****J**

- (c) Fossil fuels are often contaminated with sulfur.

State and explain why supplies of fossil fuels that contain sulfur pose a problem to the environment.

[2]

44. O/N 15/P21/Q2(c)

- (c) One of the major uses of ammonia is in the manufacture of fertilisers such as diammonium hydrogen phosphate, $(\text{NH}_4)_2\text{HPO}_4$.

- (i) Write an equation for the formation of diammonium hydrogen phosphate by the reaction between ammonia and phosphoric acid, H_3PO_4 .

[1]

- (ii) Explain this reaction in terms of the Brønsted-Lowry theory.

[2]

45. O/N 15/P22/Q3(c)

Context

- 3 The elements in Period 3, Na, Mg, Al, P and S, all react with oxygen when heated in air.

- (c) Explain how the presence of an impurity in carbonaceous fuels can give rise to acid rain.

name of impurity

[2]

46. O/N 15/P23/Q2(c)

- (c) One of the major uses of ammonia is in the manufacture of fertilisers such as diammonium hydrogen phosphate, $(\text{NH}_4)_2\text{HPO}_4$.

- (i) Write an equation for the formation of diammonium hydrogen phosphate by the reaction between ammonia and phosphoric acid, H_3PO_4 .

[1]

- (ii) Explain this reaction in terms of the Brønsted-Lowry theory.

[2]

Answers Section

1. M/J 26/P21/Q2(a)

2(a)	$\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$	1	BOD $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$ BOD $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2 \text{H}_2\text{SO}_4$
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2. M/J 26/P21/Q2(c)(i)

2(c)(i)	acid rain	1	A breathing / respiratory problems / asthma A harmful to fish / plants / animals A damages buildings / statues / trees BOD (photochemical) smog I ozone I greenhouse effect / global warming / climate change I harmful to health
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3. M/J 26/P24/Q2(a)(ii)

2(a)(ii)	M1 (N_2 molecules / N_2) have a strong triple (covalent) bond OR $\text{N} \equiv \text{N}$ (bond) is strong M2 (N_2 molecules / N_2) are non-polar	2	M1 A strong = a lot of energy to break / high activation energy / high bond energy BOD hard/difficult to break I three strong (covalent) bonds (alone) A three strong bonds between N atoms R three strong bonds AND triple bond ($\therefore = 9$) M2 A same electronegativity / no difference in electronegativity
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4. MAR 26/P22/Q2(c)

2(c)(i)	lightning	1
2(c)(ii)	PAN / peroxyac(et)lynitrate	1
2(c)(iii)	M1 first equation $\text{NO} + \frac{1}{2}\text{O}_2 \rightarrow \text{NO}_2$ M2 second equation $\text{SO}_2 + \text{NO}_2 \rightarrow \text{SO}_3 + \text{NO}$	2

5. M/J 25/P23/Q3(a)(ii)

3(a)(ii)	- unburned hydrocarbons - react with NO / NO_2 (to) produce PAN (– a component of photochemical smog) Any 2 points = 1 mark, all 3 points = 2 marks	2
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6. O/N 24/P22/Q3(a)(i)-(iii)

3(a)(i)	(natural =)lightning	1
	(man-made =)internal combustion engines	1
3(a)(ii)	$2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$ OR $4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$	1
3(a)(iii)	It / NO_2 reacts with (unburned) hydrocarbons / VOCs ALLOW reaction of unburned hydrocarbons / VOCs in presence of NO_2	1

7. O/N 24/P22/Q3(b)(iii)

3(b)(iii)	(NH_3 is an) H^+ acceptor	1
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8. O/N 24/P22/Q3(f)(ii)

3(f)(ii)	$P \equiv P$ is much weaker so P_2 is more reactive (than N_2)	1
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9. M/J 24/P23/Q1(a)(i)

1(a)(i)	M1 (N_2 molecules /molecule(s) of N_2) have a strong triple (covalent) bond M2 (N_2 molecules /molecule(s) of N_2) are non-polar	2
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10. M/J 24/P23/Q2(d)

2(d)(i)	M1 catalyst M2 2 equations to show the catalytic activity of NO_2 in formation of SO_3 from SO_2 $SO_2 + NO_2 \rightarrow NO + SO_3$ $2NO + O_2 \rightarrow 2NO_2$ M3 describe production of H_2SO_4 from SO_3 and water (can be shown in an equation). SO_3 then reacts with (rain)water / moisture / water vapour to make sulfuric acid / H_2SO_4 OR $SO_3 + H_2O \rightarrow H_2SO_4$	3
2(d)(ii)	M1 unburned hydrocarbons combine with NO and / NO_2 M2 to form PAN / peroxyac(et)yl nitrate	2

11. M/J 24/P23/Q4(b)(ii)

4(b)(ii)	$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$	1
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12. MAR 24/P22/Q3(a)

3(a)	strong triple bond / high triple bond enthalpy	1
	non-polar (molecule)	1

13. MAR 24/P22/Q3(b)

3(b)(i)	e.g. $N_2 + 1\frac{1}{2}O_2 \rightarrow NO + NO_2$	1
3(b)(ii)	HNO_2 and HNO_3	1
3(b)(iii)	photochemical smog	1

14. O/N 23/P22/Q2(d)

2(d)	natural process: lightning	1
	man-made process: internal combustion engines	1

15. O/N 23/P22/Q2(e)(ii)-(iii)

2(e)(ii)	$NO_2 + SO_2 \rightarrow SO_3 + NO$	1
	$NO + \frac{1}{2}O_2 \rightarrow NO_2$	1
2(e)(iii)	(formation of) acid rain	1

16. M/J 23/P21/Q5(b)

5(b)(i)	M1 • high temperature • high pressure • N_2 Any two correct for M1 M2 $N_2 + O_2 \rightarrow 2NO$ OR $N_2 + 2O_2 \rightarrow 2NO_2$ OR $N_2 + xO_2 \rightarrow 2NO_x$	2
5(b)(ii)	M1 NO_2 is the catalyst (for $SO_2 \rightarrow SO_3$) OR catalytic oxidation (of SO_2) by NO_2 M2 $NO_2 + SO_2 \rightarrow NO + SO_3$ AND $2NO + O_2 \rightarrow 2NO_2$ M3 SO_3 reacts with rain OR moisture/water (in atmosphere) to form sulfuric acid / H_2SO_4 OR $SO_3 + H_2O \rightarrow H_2SO_4$	3
5(b)(iii)	photochemical smog / PAN	1

17. O/N 22/P22/Q1(c)(ii)

1(c)(ii)	<i>ionic equation</i> $\text{NH}_4^+ + \text{OH}^- \rightarrow \text{NH}_3 + \text{H}_2\text{O}$	1
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18. M/J 22/P21/Q2(b)

2(b)(i)	M1 react with (unburnt) hydrocarbons M2 (form) PAN / peroxyac(et)yl nitrate	2
2(b)(ii)	$2\text{NO} + 2\text{CO} \rightarrow 2\text{CO}_2 + \text{N}_2$ OR $\text{NO}_2 + 2\text{CO} \rightarrow \frac{1}{2}\text{N}_2 + 2\text{CO}_2$	1

19. M/J 22/P21/Q2(c)

2(c)	any Group 1 hydroxide or $\text{Ca}(\text{OH})_2$ / $\text{Sr}(\text{OH})_2$ / $\text{Ba}(\text{OH})_2$	1
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20. MAR 22/P22/Q2(d)(i)-(ii)

2(d)(i)	$\text{NO} + \frac{1}{2}\text{O}_2 \rightarrow \text{NO}_2$	1
2(d)(ii)	peroxyac(et)ynitrate / PAN	1

21. MAR 21/P22/Q3(a)(ii)

3(a)(ii)	strong triple bond / high activation energy	1
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22. O/N 20/P21/Q2(b)(iii)

2(b)(iii)	(homogeneous) catalyst	1
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23. O/N 20/P23/Q2(b)(iii)

2(b)(iii)	(homogeneous) catalyst	1
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24. M/J 20/P21/Q4(a)

4(a)	Accepts a proton / H^+ (ion)	1
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25. O/N 19/P22/Q1(c)(ii)

1(c)(ii)	M1: (causes) acid rain OR reacts/dissolves with (rain)water (vapour) to form (sulfuric / sulfurous) acid M2: one point from the following list. - lowers pH / increases acidity of rivers / lakes / oceans / water supplies / seas / soil / ground water - kills/harms / damages fish - kills / harms / damages plants / damages coral / aquatic life / plants / crops / trees or deforestation - leaches (toxic) aluminium (ions / salts) from soil (into rivers / lakes) - leaches away soil nutrients / soil unfit for agriculture - damages / weathers / erodes / destroys buildings / statues	2
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26. O/N 19/P22/Q3(b)(i)

3(b)(i)	accepts a proton / H^+ OR donates a (lone) pair of e^-	1
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27. M/J 19/P23/Q4(a)

4(a)	M1 sulfur impurities OR sulfur in fossil fuels M2 converted into SO ₂ by combustion / burning sulfur OR heat sulfur with oxygen (from the air)	2
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28. M/J 19/P23/Q4(b)(ii)-(iv)

4(b)(ii)	high (enough) temperature / (a lot of) heat (energy) is produced AND to break (strong) triple bond in N ₂ / break N≡N AND nitrogen (and oxygen) from the air / atmosphere react Award two marks for three correct points Award one mark for two correct points	2
4(b)(iii)	lightning	1
4(b)(iv)	M1 nitrogen dioxide increases the rate OR lowers the activation energy M2 (NO ₂) is regenerated by reaction of NO with O ₂ (in the air) OR NO (formed) reacts with O ₂ (in air) to (re)form NO ₂	2

29. MAR 19/P22/Q1(a)

1(a)	strong triple bond / strong N≡N OR high activation energy / E _a OR non-polar	1
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30. MAR 19/P22/Q1(c)

1(c)(i)	(it is used to make) fertilisers	1
1(c)(ii)	M1 CaO displaces NH ₃ (from its salt / NH ₄ ⁺) M2 CaO is a stronger base / more basic (than NH ₃)	2

31. O/N 18/P21/Q1(c)(i)

1(c)(i)	M1 <u>acid rain</u> M2 - destroys / damages / weathers / erodes / buildings / statues - kills/harms fish / coral / plants / crops / trees / deforestation - leaches salts / ions (aluminium) from soil (into rivers / lakes) - leaches away soil nutrients - breathing difficulties - lowers pH / increases acidity of soil / rivers / oceans / seas	2
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32. O/N 18/P21/Q2(a)(i)

2(a)(i)	1 mark for each bullet, max 2 - triple bond - non-polar / no dipole - needs a lot of energy to break / strong	2
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33. O/N 18/P21/Q2(b)

2(b)(i)	(lightning) provides the (high) activation energy	1
2(b)(ii)	M1 NO + ½ O ₂ → NO ₂ M2 2NO ₂ + H ₂ O + ½ O ₂ → 2HNO ₃	2

34. O/N 18/P21/Q2(d)(ii)

2(d)(ii)	M1 ammonia / NH_3 M2 displaces NH_3	2
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35. O/N 18/P22/Q2(e)

2(e)	M1 acid rain [1] M2 any of the following [1] - lowers pH / increases acidity of rivers / lakes / oceans / water supplies / seas / soil / ground water - kills / harms / damages fish / coral / aquatic life / plants / crops / trees or deforestation - leaches (toxic) aluminium (ions / salts) from soil (into rivers/lakes) - leaches away soil nutrients / soil unfit for agriculture - damages / weathers / erodes / destroys buildings / statues - causes breathing difficulties	2
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36. O/N 18/P23/Q1(c)(i)

1(c)(i)	M1 <u>acid rain</u> M2 - destroys / damages / weathers / erodes / buildings / statues - kills/harms fish / coral / plants / crops / trees / deforestation - leaches salts / ions (aluminium) from soil (into rivers / lakes) - leaches away soil nutrients - breathing difficulties - lowers pH / increases acidity of soil / rivers / oceans / seas	2
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37. O/N 18/P23/Q2(a)(i)

2(a)(i)	1 mark for each bullet, max 2 - triple bond - non-polar / no dipole - needs a lot of energy to break / strong	2
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38. O/N 18/P23/Q2(b)

2(b)(i)	(lightning) provides the (high) activation energy	1
2(b)(ii)	M1 $\text{NO} + \frac{1}{2} \text{O}_2 \rightarrow \text{NO}_2$ M2 $2\text{NO}_2 + \text{H}_2\text{O} + \frac{1}{2} \text{O}_2 \rightarrow 2\text{HNO}_3$	2

39. O/N 18/P23/Q2(d)(ii)

2(d)(ii)	M1 ammonia / NH_3 M2 displaces NH_3	2
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40. M/J 18/P22/Q3(d)(ii)-(iii)

3(d)(ii)	$\text{NO}_2 + \text{SO}_2 \rightarrow \text{NO} + \text{SO}_3$	1
	$\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$	1
3(d)(iii)	(it is) regenerated / not used up / undergoes temporary chemical change / recovered unchanged	1
	$\text{NO} + \frac{1}{2} \text{O}_2 \rightarrow \text{NO}_2$	1

41. M/J 17/P22/Q2(a)

2(a)	strong triple bond	1
	non-polar / no dipole	1

42. M/J 17/P22/Q2(b)

2(b)(i)	Any 2 points covered correctly scores 2 marks Any 1 point covered correctly scores 1 mark - nitrogen (and oxygen) from the air / atmosphere (react): - high temperature (of internal combustion engine) / (engine) produces enough OR a lot of heat (energy) : (so) breaks (strong) bond(s) in nitrogen (and oxygen) :	2
2(b)(ii)	reduction / decomposition of NO _x using a catalyst / catalytic convertor 2NO ₂ + 4CO → 4CO ₂ + N ₂ OR 2NO + 2CO → 2CO ₂ + N ₂	1
2(b)(iii)	(acts as a homogeneous) catalyst OR oxidising agent SO ₂ + NO ₂ → SO ₃ + NO NO + ½O ₂ → NO ₂ OR SO ₃ + H ₂ O → H ₂ SO ₄	1
2(b)(iv)	2NO ₂ + H ₂ O → HNO ₂ + HNO ₃ OR 4NO ₂ + 2H ₂ O + O ₂ → 4HNO ₃	1

43. MAR 16/P22/Q4(c)

(c)	sulfur dioxide would be produced on combustion (which contributes to) <u>acid rain</u>	[1] [1]	[2]
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44. O/N 15/P21/Q2(c)

(c) (i)	2NH ₃ + H ₃ PO ₄ → (NH ₄) ₂ HPO ₄	[1]	[1]
(ii)	NH ₃ identified as base AND H ₃ PO ₄ identified as acid base accepts protons AND acid donates protons	[1] [1]	[2]

45. O/N 15/P22/Q3(c)

(c)	sulfur forms SO ₂ /SO ₂ + /mixes H ₂ O → H ₂ SO ₃ or in words OR SO ₂ + /mixes H ₂ O (→acid) / or in words OR SO ₂ + /mixes H ₂ O + (1/2O ₂) →H ₂ SO ₄ / or in words	[1] [1]	[2]
			[14]

46. O/N 15/P23/Q2(c)

(c) (i)	2NH ₃ + H ₃ PO ₄ → (NH ₄) ₂ HPO ₄	[1]	[1]
(ii)	NH ₃ identified as base AND H ₃ PO ₄ identified as acid base accepts protons AND acid donates protons	[1] [1]	[2]